UDBHAVA-2026

FINAL PPT PRECENTATION
SC-01 THE “GHOST” CLASS ROOM

PROBLEM ⚠️

GHOST CLASSROOM ENERGY

- ▪ Classrooms remain unused due to cancellations or schedule gaps
- ▪ HVAC & lighting continue running unnecessarily
- ▪ No real-time link between timetable and energy systems
- ▪ Manual switching is slow, inconsistent, and inefficient
- ▪ ~40% energy is wasted → high cost + carbon emissions

SOLUTION 💡

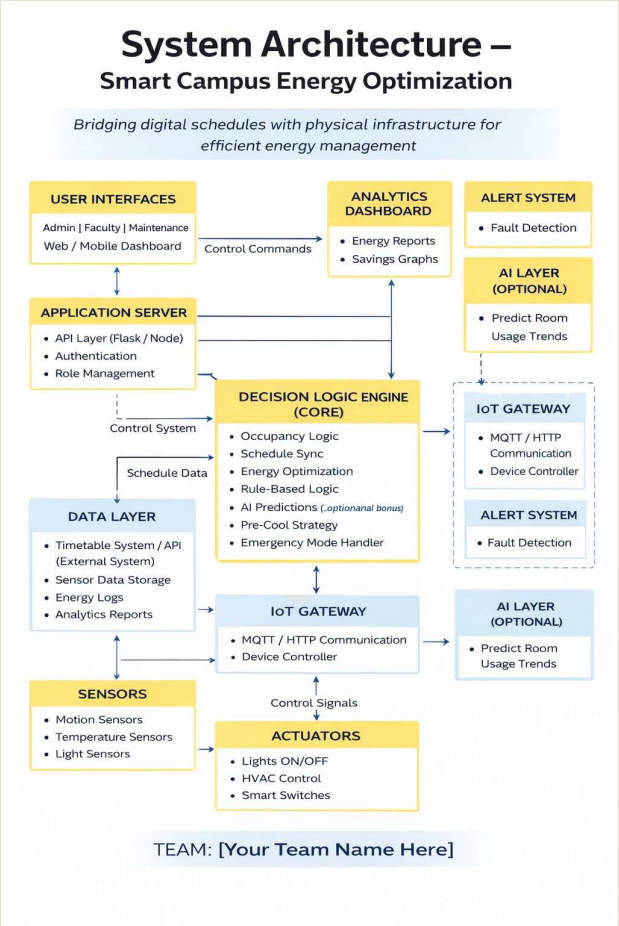
Automated Energy Grid Orchestrator

- ▪ Integrates timetable + occupancy sensors + energy systems
- ▪ Detects real-time room usage
- ▪ Automatically turns OFF systems when room is idle
- ▪ Prepares room before next class (pre-cooling / pre-heating)
- ▪ Ensures energy efficiency without affecting comfort

ARCHITECTURE 🏗️

System Design Overview

- Input Layer → Timetable Database + IoT Sensors
- Processing Layer → Decision Engine (Rules + Logic)
- Control Layer → HVAC & Lighting Systems
- Application Layer → Dashboard (Admin / Faculty)
- Communication via APIs for real-time automation
- Modular & scalable system design



DEMO ⚙️

- **System Workflow**
- ▪ Fetch timetable data continuously
- ▪ Sensors detect occupancy in real-time
- ▪ If room is idle > 15 mins → Turn OFF lights & HVAC
- ▪ Before next class → Pre-adjust temperature
- ▪ Dashboard allows monitoring & manual override

FEATURES 🚀

Key Functionalities

- Real-time occupancy detection
- Automated energy control (HVAC + Lights)
- Predictive temperature adjustment
- Manual override for faculty
- Maintenance alerts ⚠️
- Energy usage analytics 📊
- Emergency “All-ON” mode 🔴